

SHIVAM SINGH

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SUMMARY

Final-year B.Tech CSE (AI) student and software engineer with hands-on experience building scalable backend systems, cloud-native applications, and AI-powered solutions. Experienced in developing end-to-end applications spanning serverless AWS architectures, machine learning pipelines, LLM integration, and retrieval-augmented systems. Proficient in Python, Django, React, Docker, and AWS (Lambda, DynamoDB, CDK), with experience in vector databases and cloud-based ML services. AWS and Oracle certified in Machine Learning and Generative AI.

EDUCATION

B.Tech, Computer Science Engineering (Artificial Intelligence)

2023 – 2027

Intermediate: 84.17% | High School: 95%

TECHNICAL SKILLS

Languages: Python, Java, SQL, C, HTML/CSS, TypeScript

Frameworks & Libraries: Django REST Framework, React + Vite, scikit-learn, Pandas, NumPy, Matplotlib, TailwindCSS, shadcn/ui

AI / ML: Machine Learning, Deep Learning, NLP, RAG, LLM Integration (Ollama/Qwen2.5), Generative AI, Feature Engineering

Databases & Vector DBs: PostgreSQL, Qdrant, Redis, SQL

Cloud & DevOps: AWS (Lambda, DynamoDB, S3, CDK, Rekognition, SageMaker A2I, Cognito, SNS/SQS, CloudWatch), Oracle Cloud Infrastructure, Docker, Docker Compose, Celery, GitHub Actions

PROJECTS

HITL Content Moderation Engine – Human-in-the-Loop AWS Pipeline

Stack: AWS CDK (Python), Lambda, S3, DynamoDB, Amazon Rekognition, SageMaker Ground Truth (A2I), Cognito, SNS, SQS, Streamlit

- Architected a serverless, event-driven content moderation pipeline on AWS CDK that routes uploaded images through Rekognition-based confidence scoring into three tiers — auto-approve, human review, and auto-quarantine — backed by a DynamoDB status table with GSI-based queue querying.
- Built a human-in-the-loop reviewer portal (Streamlit + Cognito auth) integrated with SageMaker Ground Truth A2I for ambiguous-confidence cases, and hardened the pipeline for production with dead-letter queues, retry policies, SNS alerting, and least-privilege IAM.

NotyBrain – AI-Powered Personal Knowledge Management System

Stack: React + Vite + TypeScript, Django REST Framework, PostgreSQL, Qdrant, Celery + Redis, Ollama (Qwen2.5), Docker

- Architected a full-stack, privacy-first knowledge management SPA with a RAG pipeline backed by a locally-run LLM (Qwen2.5 via Ollama), enabling multi-format content ingestion with zero data leakage to third-party APIs.
- Built semantic vector search (Qdrant + sentence-transformers), a grounded Q&A engine with source citations to eliminate hallucinations, and a Knowledge Graph API for D3.js/Cytoscape visualization.

Drone-Based AI System for Deforestation Pattern Monitoring

Stack: Python, Computer Vision, NDVI Analysis, Geospatial Data Processing, Deep Learning

- Designed an AI-powered remote sensing pipeline using NDVI analysis and geospatial data to detect and quantify deforestation patterns from satellite imagery across multiple temporal observations.

AI-Powered Driver Fatigue Drift Monitoring System

Stack: Python, PyTorch, MediaPipe FaceMesh, OpenCV, Streamlit, Captum, ONNX, Docker

- Developed a real-time fatigue monitoring framework that models driver fatigue as a continuous temporal drift process using BiLSTM-GRU architecture, weakly supervised Multiple Instance Learning, and personalized baseline calibration.
- Built a sub-20 ms multi-threaded inference pipeline with explainable AI attributions, changepoint detection, confidence estimation, privacy-preserving processing, and an interactive monitoring dashboard for real-time fatigue tracking.

CERTIFICATIONS

AWS Machine Learning Foundations | AWS Agentic AI | Oracle OCI Generative AI Professional | Oracle OCI AI Foundations